To make this book enjoyable to a wide range of readers, I have used the concise style standard in scientific references and have avoided topics of only scholarly interest. Readers who wish to dig deeper can find an extended bibliography, additional information, and certain scholastic qualifications on the website www.creatingtechnology.org.

1

INTRODUCTION

"The engineer typifies the twentieth century. Without his genius and the vast contributions he has made in design, engineering, and production on the material side of our existence, our contemporary life could never have reached its present standard." Thus wrote Alfred Sloan, who, with a college degree in engineering, climbed the white-collar career path to the presidency of General Motors. Engineering not only elevates society's material culture but also makes for satisfying personal lives, as observed by Herbert Hoover, a mining engineer whose life career culminated as president of the United States: "It is a great profession. There is the fascination of watching a figment of the imagination emerge through the aid of science to a plan on paper. Then it moves to realization in stone or metal or energy. Then it brings jobs and homes to men. Then it elevates the standards of living and adds to the comforts of life. That is the engineer's high privilege."[1]

Engineering is the art and science of production that, alongside reproduction, is the most fundamental of human activities. As practical art, it flourished with the rise of civilization. Even in ruins, monuments around the world display the ingenuity of ancient master builders. Modern engineering, which emerged together with modern science in the seventeenth-century scientific revolution, amplifies traditional ingenuity by the power of scientific reasoning and knowledge. Directly in-

volved in material production, it acts at the vortex, merging research and development, on the one hand, and industry and business, on the other. Science, engineering, and business jointly constitute the main engines of technology, whose products have been woven into the fabrics of our daily life. This book focuses on engineering, not neglecting its overlaps and bonds with the other two. What are the characteristics of engineering thinking and practice that make them so effective in advancing technology? How are they related to those of natural science? What factors do engineers reckon with in designing a system? How do their decisions shape technological evolution?

As a creative and scientific activity that transforms nature to serve the needs and wants of large numbers of people, engineering has both physical and human dimensions. To modify nature effectively as desired requires mastery of natural laws and phenomena, thus engineering shares the contents and standards of natural science. To ascertain what modifications are desirable requires an understanding of human and socioeconomic factors, thus engineering goes beyond natural science in its missions of utility and service. Engineers address things and people, bringing nature and humanity together. They develop physical technology to grasp general principles, extract materials, create implements, design products, and invent processes for efficient manufacturing and operation. They also develop organizational technology to analyze goals, assess costs and benefits, trade off factors under constraints, negotiate consensus among involved parties, plan projects, and coordinate human efforts in production. The physical and organizational dimensions infiltrate and reinforce each other in the advancement of engineering, technology, and social well-being.

Engineering has progressed remarkably during the past century. Its scientific sophistication soars with the complexity of technologies it creates. Its management and social horizon expand with the scope of production and technological risks. Exercising the physical and human arms of their profession, engineers engage in science, design, and leadership. Through these activities they improve knowledge, implements, and organizations, which are the major repositories of technology, society's scientific productive capacity.

Engineering is scientific. Its first aspect, science, is closely related but not identical to natural science. The relation between the two was aptly

described by electrical engineer Vannevar Bush, whose 1945 report to the U.S. president, entitled *Science—The Endless Frontier,* set the vision of the National Science Foundation (NSF). He wrote elsewhere: "In all associations between engineers and scientists, engineering is more a partner than a child of science . . . While everyone knows that engineering is concerned with the conversion of science into technology, everyone does not know that engineering also does just the opposite and translates technology into new science and mathematics."[2] Scientists know. Physicist James Clerk Maxwell acknowledged in his treatise that revolutionized electromagnetic theory: "The important applications of electromagnetism to telegraphy have also reacted upon pure science by giving a commercial value to accurate electrical measurements, and by affording to electricians the use of apparatus on a scale which greatly transcends that of any ordinary laboratory. The consequences of this demand for electrical knowledge, and of these experimental opportunities for acquiring it, have been already very great, both in stimulating the energies of advanced electricians, and in diffusing among practical men a degree of accurate knowledge which is likely to conduce to the general scientific progress of the whole engineering profession."[3] That was 1873. Since then the partnership of science and engineering has strengthened so tremendously that science and engineering are often treated as a unit by the NSF.

Natural scientists discover what was not known. Engineers create what did not exist. Both boldly go where no one has gone before, each original and creative in its own way. Through intensive research efforts in the past fifty years, engineers have developed engineering sciences, bodies of coherent knowledge that are comparable to the natural sciences in their length of vision, breadth of scope, depth of analysis, level of creativity, rigor of research, and criteria of acceptance. In the process they are neither imitating natural scientists nor ignoring them. Engineers mainly develop and refine the independent reasoning inherent in productive activities, which both Aristotle and Galileo analyzed and treated on equal footing as reasoning in pure study. Independence is not insularity. Engineers and natural scientists may have different motivations and concerns, but they share the human mind and the physical world. They have common knowledge, methods, and ways of thinking, which include mathematics and instrumentation, theory formation

and controlled experimentation, research and development. In many workplaces they collaborate so closely it is difficult to tell them apart. Their domains increasingly overlap as swelling knowledge bursts the boundaries of academic departments and society increasingly demands useful results from research. Genetic engineering occupies much of cutting-edge biology, and nanotechnology looms as a direction for future chemistry. Convergence accelerates and interdisciplinary research centers mushroom as the twenty-first century dawns.

While engineering science expands knowledge, design creates specific implements and systems, ranging from the goods and services that people use every day to society's infrastructures for energy, communication, transportation, public health, and national defense. Design is not unique to engineering, but its paramount importance is. Design of good experiments is crucial to natural science, but there it is at the service of the central goal of acquiring knowledge. The emphasis is reversed in engineering, where design is the central goal that calls on the service of scientific knowledge. Engineering design is creative and imaginative, although invariably constrained by many conditions that the intended product is required to meet. It includes processes from the conception of a system through development to the plan for producing it. In the process engineers must attend to the performance of the system as a whole as well as its components down to the smallest detail, not neglecting the erratic environment in which it is intended to operate. To ensure that the product works reliably and safely, they try to anticipate problems and head them off. No scientific principle can cover all the vicissitudes of reality. Thus design engineers must complement their mathematical prowess and explicit knowledge with practical insight and know-how. Insights and experiences originated in design practice, when scrutinized, developed, and systematized, become new scientific engineering knowledge. In his seminal study in the epistemology of design, aeronautic engineer Walter Vincenti remarks that a great number of engineers "engage in design, and it is there that requirements for much engineering knowledge originate in an immediately technical sense."[4]

Design is akin to invention. In creating technology, however, engineers do not merely invent, they innovate. Innovation, which includes the whole process from discerning the need for an invention to bringing

the new idea through design and production to large-scale use and economic return, has a broad perspective that requires both technical and social acumen. Consider designing and building a subway system, for instance. Even when funding is available, the communities to be served—or inconvenienced—are heterogeneous populations, not to mention the politicians, environmentalists, and myriad other interest groups involved. Engineers must listen to and negotiate with them all, and the outcomes will influence the design of the system, such as the locations of stations. Then the construction process brings in more parties, from workers and contractors to drivers cursing traffic detours. The importance of organizational technology is unmistakable.

Engineered systems are real and have real consequences. Engineers have heavy social responsibilities to manage intricate technologies that, while elevating living conditions, also carry risks for accidents, abuses, and harmful side effects. To meet the challenge, they expand their horizon to cover societal needs, government regulations, and impacts on the environment. It falls on them to penetrate media hype and ideological rhetoric to explain realistic factors clearly to the public, thus to help people to evaluate relative importance, weigh risks, make trade-offs, and decide rationally on public policies concerning technology. For these tasks engineering schools are intensifying education in social and communicative skills. Many engineers agree with their colleague Norman Augustine: "Engineers must become as adept in dealing with societal and political forces as they are with gravitational and electromagnetic forces."[5]

The dual physical and human dimensions and triple aspects of science, design, and leadership provide the framework for our exploration of engineering. This multidimensional framework does not imply that each engineer is expert in all three aspects. Engineering is a large and diverse field. There are more than two million working engineers and more than a hundred professional societies in the United States tending to mechanical, information, and other kinds of technology. Overwhelming technological complexity makes the division of intellectual labor inevitable. Thus engineer-scientists focus on research, engineer-entrepreneurs and engineer-managers on organization. The majority of engineers focus on design and development, which many regard as the core of engineering. The focus, however, is not tight. An engineer work-

ing in one aspect of engineering is aware of the other two, although not with deep expertise. Engineer-scientists are not confined to ivory towers, design engineers are not mere technicians, and engineer-managers are not simply bean counters. They are all engineers concerned about creating and advancing technology. The broad view of engineering is aptly described by civil engineer Frederick Clarke:

> It involves the delicate and difficult task of translating scientific abstraction into the practical language of earthly living; and this is perhaps the most completely demanding task in the world. For it requires an understanding of *both* spheres—the pure ether in which science lives, and also the goals and drives and aspirations of human society in all its complexity. The engineer must be at once philosopher, humanist and hard-headed, hard-handed worker. He must be a philosopher enough to know what to *believe,* humanist enough to know what to *desire,* and a workman enough to know what to *do.*[6]